

Code Change Plan

Mar 30 – Apr 1, 2026

Mar 30 (Sun) — Locate Problems & Begin Yellow-Plot Fix

Today I traced through the GUI rendering pipeline and identified the root causes of the three issues.

- The AE / DA / DE heatmaps appear fully yellow in an empty room because `_update_derived_heatmaps()` normalises each map by its own max value. When the noise floor is uniform, every pixel ends up near 1.0 and maps to the yellow end of the colormap.
- Located the relevant code in `core/update_loop.py` (the `_update_derived_heatmaps` function).
- Started adjusting the AE map's `levels` parameter to use percentile-based scaling instead of `[0, 1.0]`, so that uniform low-signal frames stay blue.

Mar 31 (Mon) — RTI/RDI Stitching & Finish Yellow Fix

- Apply the same percentile-based levels fix to the DA and DE maps.
- Fix the visible “stitching” artefact in RTI and RDI plots. The cause is in `processing/dsp_main.py`: frames are accumulated in `rti_queue` / `rdi_queue` and concatenated directly, so amplitude jumps appear at frame boundaries. Plan to add per-frame normalisation before concatenation.
- Test with empty room + person present.

Apr 1 (Tue) — Point Cloud / Trajectory / Height Tabs

- Try to locate the root cause of the three non-functional tabs (Point Cloud, Trajectory, Height) and attempt initial fixes.